

Model Tuning

Agenda

- K-Fold Cross-Validation
- Oversampling and Undersampling
- Data Leakage and Data Imbalance
- Grid Search
- Randomized Search

Let's begin the discussion by answering a few questions

Model Tuning Quiz

What does a larger value of k in k -fold cross-validation typically imply in terms of computational complexity?

A

Increased training time

B

Reduced memory usage during training

C

Faster cross-validation process

D

Lower computational cost per fold

Model Tuning Quiz

What does a larger value of k in k -fold cross-validation typically imply in terms of computational complexity?

A

Increased training time

B

Reduced memory usage during training

C

Faster cross-validation process

D

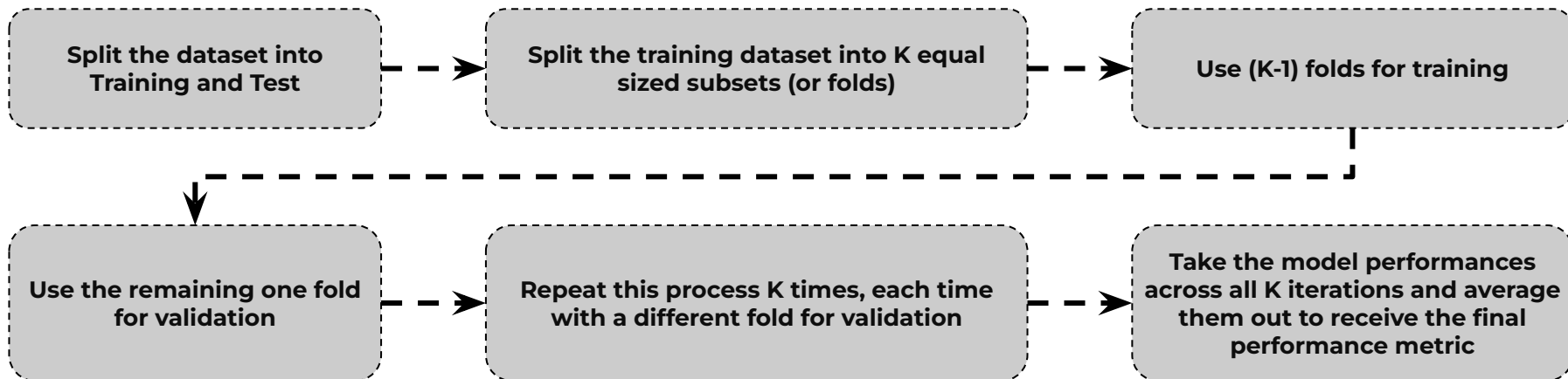
Lower computational cost per fold

K-Fold Cross Validation

A **resampling technique** used to **evaluate the performance** of an ML model

Helps assess how well a **model generalizes to unseen data**

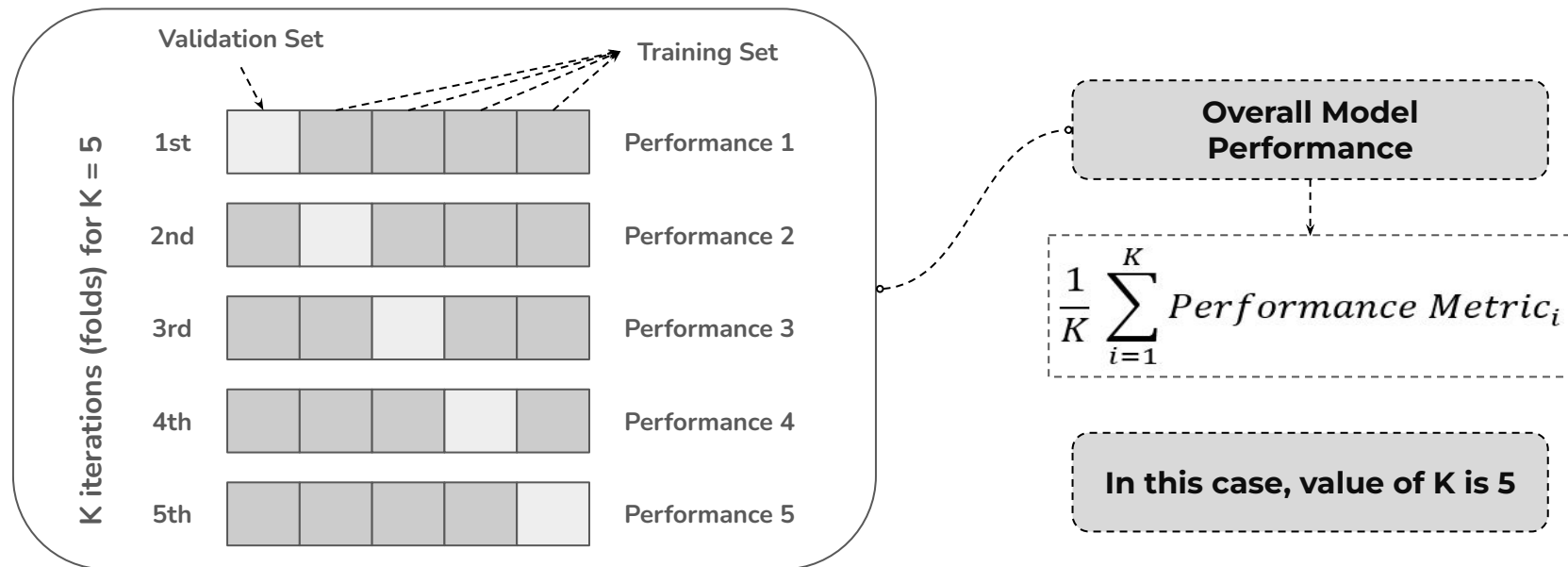
Steps for K-Fold Cross Validation



K-Fold Cross Validation

Model has to be **trained K times**, each time with a **different training set**

Higher value of K results in more training runs, and **increased training time**.



Which of the following can cause data leakage in a machine learning model?

A

Filling missing values in the training data with attribute means computed from the training data

B

Filling missing values in the test data with attribute means computed from the test data

C

Filling missing values in the test data with attribute means computed from the training data

D

Filling missing values in the training data with attribute means computed from the test data

Which of the following can cause data leakage in a machine learning model?

A

Filling missing values in the training data with attribute means computed from the training data

B

Filling missing values in the test data with attribute means computed from the test data

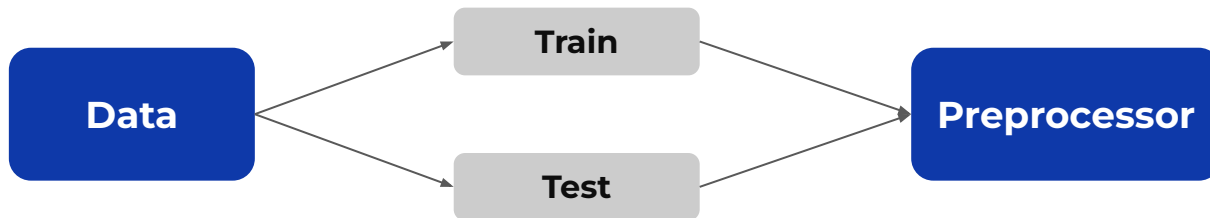
C

Filling missing values in the test data with attribute means computed from the training data

D

Filling missing values in the training data with attribute means computed from the test data

Data Leakage



If **preprocessing** (e.g., scaling, encoding, missing value imputation) is done **before splitting**, the test set will contain **information derived from the training dataset**, potentially **leading to artificially high performance**.

As the **test data is unseen data**, we **cannot use its attributes** to preprocess the training data.

Class Imbalance

Class imbalance occurs when one class has significantly more observations than the other(s).

Models tend to get biased toward the majority class, leading to misleadingly high accuracy

Minority class predictions often suffer from low recall, which is critical in real-world problems

Standard evaluation metrics may not reflect true model performance under class imbalance. Hence, class imbalance is essential to build a robust model

Model Tuning Quiz

In an imbalanced classification problem, _____ is used to increase the number of observations in the _____ class, whereas _____ is used to reduce the number of observations in the _____ class

A

Undersampling, Majority, Oversampling, Minority

B

Oversampling, Minority, Undersampling, Majority

C

Oversampling, Minority, Undersampling, Minority

D

Oversampling, Majority, Undersampling, Majority

Model Tuning Quiz

In an imbalanced classification problem, _____ is used to increase the number of observations in the _____ class, whereas _____ is used to reduce the number of observations in the _____ class

A

Undersampling, Majority, Oversampling, Minority

B

Oversampling, Minority, Undersampling, Majority

C

Oversampling, Minority, Undersampling, Minority

D

Oversampling, Majority, Undersampling, Majority

Undersampling V/S Oversampling

Undersampling

Reduces the number of samples from the majority class

Decreases overall dataset size

May lead to loss of important information

Preferred when the dataset is very large and redundant

Oversampling

Increases the number of samples in the minority class

Increases overall dataset size

May lead to overfitting due to duplicate samples

Preferred when the dataset is small and losing data is risky

Model Tuning Quiz

You are working on a machine learning project where you need to fine-tune the hyperparameters of a model to achieve the best performance. You have a dataset with a moderate size, and you are considering using a hyperparameter optimization method. Which of the following scenarios would be most appropriate for using Grid Search?

A

The computation time is not a concern, and you need to exhaustively search over all possible combinations

B

The computation time is a major concern, and you do not need to evaluate all combinations

C

You need fast results with limited computation resources

D

Perform model selection across multiple algorithms without hyperparameter tuning

Model Tuning Quiz

You are working on a machine learning project where you need to fine-tune the hyperparameters of a model to achieve the best performance. You have a dataset with a moderate size, and you are considering using a hyperparameter optimization method. Which of the following scenarios would be most appropriate for using Grid Search?

A

The computation time is not a concern, and you need to exhaustively search over all possible combinations

B

The computation time is a major concern, and you do not need to evaluate all combinations

C

You need fast results with limited computation resources

D

Perform model selection across multiple algorithms without hyperparameter tuning

Grid Search

An algorithm used to find the **optimal set of hyperparameters** for a **model** from the **provided search space**

Consider the following set of Random Forest hyperparameters and their values to check

```
{  
    "n_estimators": [50, 75, 100],  
    "max_depth": [4, 5]  
}
```

Following combinations of hyperparameters will be tried out

n_estimators	50	50	75	75	100	100
max_depth	4	5	4	5	4	5

Model Tuning Quiz

You are tuning a machine learning model using RandomizedSearchCV. You specify distributions (ranges) for each hyperparameter instead of fixed lists, and set `n_iter = 40` because full grid search would take too long. You know that:

- 1. Only 40 random combinations of hyperparameters will be tried,**
- 2. The combinations are sampled randomly(not sequentially),**
- 3. Each hyperparameter can be given as a continuous range.**

In which scenario is this use of Random Search most appropriate?

A

When you must exhaustively evaluate every possible hyperparameter combination

B

When you want to randomly sample a fixed number (`n_iter`) of hyperparameter combinations from a large search space to save time

C

When you only have 2–3 hyperparameters, each with 2 fixed values

D

When the hyperparameters cannot be represented as ranges or probability distributions

Model Tuning Quiz

You are tuning a machine learning model using RandomizedSearchCV. You specify distributions (ranges) for each hyperparameter instead of fixed lists, and set `n_iter = 40` because full grid search would take too long. You know that:

- 1. Only 40 random combinations of hyperparameters will be tried,**
- 2. The combinations are sampled randomly(not sequentially),**
- 3. Each hyperparameter can be given as a continuous range.**

In which scenario is this use of Random Search most appropriate?

A

When you must exhaustively evaluate every possible hyperparameter combination

B

When you want to randomly sample a fixed number (`n_iter`) of hyperparameter combinations from a large search space to save time

C

When you only have 2–3 hyperparameters, each with 2 fixed values

D

When the hyperparameters cannot be represented as ranges or probability distributions

Randomized Search

An algorithm that **randomly samples combinations of hyperparameters** from a defined search space to **identify an optimal or near-optimal model configuration efficiently**.

Consider the following set of Random Forest hyperparameters and their values to check

```
{  
    "n_estimators": [50, 75, 100],  
    "max_depth": [4, 5]  
}
```

Following combinations of hyperparameters will be tried out (**n_iter = 3**):

n_estimators	75	100	50
max_depth	4	5	4

Only n_iter = 3 combinations
evaluated from the full search space



Power Ahead !

